

Nathan Zimmerman

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WORK HISTORY

Software Engineer II, Nelnet

10/2023 – Present

- Contributed to a greenfield Payment Plans & Billing domain for a multi-tenant platform serving schools, built with C#/.NET microservices and Angular.
- Built a payment plan schedule calculator in the enrollment service. Generated installment scenarios based on first-installment constraints, installment count, down payment, frequency (weekly/monthly), and completion date; wired results to an interactive Angular UI (sliders/inputs).
- Packaged the calculator as a npm package and integrated it into the payment plans service for admin preview.
- Subscribed to payments service for returned payments and added handlers in payment plans service to apply return policy actions (e.g., rescheduling installments or cancelling plans).
- Integrated with an internal correspondence service to trigger automated enrollment confirmation emails.
- Implemented runtime payment plan status calculation from persisted confirmation/start/end timestamps to keep UI state accurate without storing status in the database.
- Improved data quality by expanding FluentValidation rules and adding unit/integration tests xUnit and Moq.

Software Development Engineer, Amazon

05/2022 – 04/2023

- Engineered risk guidance applications using Java, Scala, Spring, and AWS to screen third-party sellers against denied party lists, predicting financial offenses such as money laundering.
- Remediated critical security issues by enabling encryption (SQS/SNS/S3 via KMS), tightening IAM policies to least privilege, and upgrading dependencies to address vulnerabilities.
- Performed EC2 host migrations across IAD/PDX Availability Zones for 10+ applications and 100+ environments, validating cutovers with alarms/dashboards to ensure zero downtime.
- Reduced support ticket volume by resolving pipeline failures, monitoring alarms, and fixing deployment errors.

Software Developer, Radian Group Inc.

09/2019 – 04/2022

- Contributed to launch and maintenance of “MyRadian,” a React + Node.js suite for Valuation and Asset Management Services, supporting feature delivery through code reviews, bug fixes, and production support.
- Troubleshoot production issues and improved performance using AWS (CloudWatch, X-Ray, DynamoDB); maintained legacy C#/.NET and Java apps, resolving customer issues with Splunk and SQL.
- Automated database reconciliation workflows with Python scripts, eliminating manual daily effort.

PROJECTS

- [Sudoku](http://sudoku.nathanzimmerman.com) (sudoku.nathanzimmerman.com) - Built a full-stack Sudoku app with a .NET Web API puzzle generator and Angular gameplay UI. Added validation, rate limiting, and CI-backed tests to keep gameplay reliable.
- [Blackjack](http://blackjack.nathanzimmerman.com) (blackjack.nathanzimmerman.com) - Built a session-backed Blackjack game with Spring Boot + Spring Security and a React UI. Implemented rules (splits, insurance) plus validation and security hardening.
- [Nerdle](http://nerdle.nathanzimmerman.com) (nerdle.nathanzimmerman.com) - Built a Wordle-style game with a Node/Express backend that validates guesses server-side. Added rate limiting, theming, and stats tracking.
- [Brick Breaker Resume](http://resume.nathanzimmerman.com) (resume.nathanzimmerman.com) - Built a Phaser game that parses a live .docx resume to generate levels from text. Implemented mapping logic, responsive scaling, and score persistence.

SKILLS

- HTML, CSS, JavaScript, Angular, React, Java, Node.js, C#, .NET, MongoDB, Azure, AWS, REST API, Git, Claude

EDUCATION

AS, Computer Science, Delaware County Community College, 2018

BA, Psychology, (History minor), Rutgers University School of Arts and Sciences, 2016